$$|\Psi\rangle = \sum_j \alpha_j |\psi_j\rangle,$$

where the $|\psi_j\rangle$ are the orthonormal eigenvectors of a Hermitian operator, the collection of coefficients α_j—the things we called $\psi(a,b,c,\ldots)$ just above—is what we mean by the wave function. In situations where the state-vector is expressed as an integral rather than a sum, the wave function is continuous rather than discrete.

So far, we have been careful to distinguish the wave function from the state-vectors $|\psi\rangle$, and this is a common convention. However, some authors refer to wave functions as though *they* are the state-vectors. This ambiguous use of terminology can be confusing. It becomes less confusing when you realize that a wave function really can represent a state-vector. It is reasonable to think of the α_j coefficients as the coordinates of the state-vector in a specific basis of eigenvectors. This is similar to saying that a set of Cartesian coordinates represents a particular point in 3-space relative to a specific coordinate frame. To avoid confusion, just try to be aware of which convention is being followed. In this book, we will generally use uppercase symbols, such as Ψ, to represent state-vectors, and lowercase symbols, such as ψ, to represent wave functions.

5.2 MEASUREMENT

Let's return to the concept of measurement. Suppose we measure two observables **L** and **M** in a single experiment, and the system is left in a simultaneous eigenvector of these two observables. As we learned in Section 5.1.1, this means that **L** and **M** must commute.

But what if they don't commute? Then, in general, it is not possible to have unambiguous knowledge of both. Later on, we will make this more quantitative in the form of the uncertainty principle, Heisenberg's being a special case.